

KIT502 Web Development

Group Assignment for Developing Web Application

Subject	KIT502 Web Development
Assessment Title	Full-Stack Web Application Development
Assessment Type	Group
Weighting	45% total (Assignment 1: 15%, Assignment 2: 30%)
Due	Assignment 1 Week 5 August 5 Assignment 2 Week 12 September 30
Intended Learning outcome	ILO1, ILO2, ILO3

Assignment Overview

A real estate company has engaged your development team to design and develop a modern **Real Estate Management Platform**. The application will be implemented using the Laravel framework with MySQL (or SQLite) as the database backend, following the Model–View–Controller (MVC) architecture.

The client requires a responsive, secure, and database-driven web application that allows users to browse, buy, rent, and manage property listings through an online platform.

The project will be developed across two stages:

- **Assignment 1: Front-End Prototype and Database Design**
Students will design and implement the user interface prototype and develop the database design required to support the system.
- **Assignment 2: Full-Stack Laravel Application**
Students will extend Assignment 1 by implementing a complete Laravel application with database integration, user authentication, authorization, and functional system features.

Throughout the project, students are expected to apply industry-standard web development practices, including responsive design, usability principles, database modelling, authentication and authorization, CRUD operations, and secure web application development practices.

Development Environment

All project files must be stored in the designated group directory:
`/groupwork/kit502-group-##`

on the `ictteach-usermin` (Melbourne and Ultimo: `192.168.200.4/`) server, where `##` represents the final two digits of your MyLO group number.

Files stored outside the assigned group directory may not be considered for assessment.

Assignment 1 – Front-End Web Application and Database Design

Develop a fully functional front-end prototype demonstrating the appearance and client-side behaviour of the proposed real estate platform.

No server-side programming is required in this assignment.

Students should carefully analyse the system requirements and prepare an appropriate database design that will support Assignment 2.

Site navigation: The navigation menu must

- appear consistently across every page
- contain links to all pages
- indicate the active page
- include hover/focus effects
- remain fixed (sticky navigation)
- collapse appropriately on mobile devices

Landing page: This page serves as the landing page for real estate platform. It is intended to be the first page your site visitors will see.

- Hero banner¹
- Featured properties (minimum 3)
- Property search section (front-end only)
- Testimonials (minimum 2)
- Company highlights
- Footer

Registration page: This is a page where a user registers into your website. This registration page should include a form with input fields of role (buyer or seller), name, email address, password, and confirmed password. It should include register and reset button.

- Include JavaScript/JQuery field validation to prevent the registration form from being submitted if:
 - Any of the fields are empty
 - The email address field does not contain a valid email format (i.e., it must be well-formed, but you are not required to verify whether the email address exists).
 - The password and confirm password fields do not match (i.e. do not contain the same value)
 - The password does not conform to a password policy
 - Password policy: a minimum of 7 characters and at least one upper case and lowercase and one special character.
- Clicking the reset button should clear all form fields
- Submitting the form should take the user back to the landing page.

¹ The hero banner area is the area of the page between the navigation and the start of your content. The hero banner is often an image, but it doesn't have to be. It serves as the starting point for your page and its contents should orient the user to the content on the rest of the page.

Login page: This page will allow users of the platform to login to gain access to member-only area.

- The page should include a form with input and label tags for an email, a password, and a login button.
- Include JavaScript/jQuery field validation to prevent the form being submitted if any of fields are empty.
- Submitting the form should take the user back to the landing page.

Property listing Page: This page will list the property

- Display at least 6 properties, responsive grid layout
- Search box
- Filter Panel: it should filter by property type, bedrooms and status
- Sorting by price, bedrooms
- Link to property detail page

Note: Filtering and sorting can be implemented using JavaScript or jQuery with locally stored data (no database).

Property Details Page: this page displays the details of the property.

- Image gallery
- Property description
- Property specifications (price, type, bedroom numbers, bathroom numbers, size of the place, number of parking spots available)
- Seller's information

About Page: this page displays the information of real estate company

- Company overview
- Team members
- Mission and visions

Contact Page: this page displays the contact information of the company

- Contact form (name, email and description)
- Company contact information

Database Design: you are going to design a schema for a database that you will use in assignment 2. You are required to produce an ER diagram / database schema. It should include:

- Entities, attributes (including primary and foreign keys) and relationships
- Appropriate data types for your attribute.

You may use the free version of Lucidchart or Draw.io to draw ER diagrams, although it is not the requirement. Hand-drawn diagrams are equally good. Submit your database schema and ER diagram in a PDF file to MyLO.

ReadMe File: It provides details about your group and your website, includes each member's contribution, any external resources used and presents an assignment progress plan (e.g, work allocation among group members, timeline plan, etc).

Submit the Coding Deliverables to the icteach-usermin Server

- HTML files
- CSS files
- JavaScript/jQuery files
- Images

Submit the Remaining Deliverables to MyLO

- Database Schema / ER diagram
- ReadMe file

The **Self and Peer Evaluation** must be completed by **Friday, 7 August**.

Assignment 2: Completed Full-Stack Web application

Extend Assignment 1 into a complete Laravel web application using MySQL or SQLite as the database backend.

Students are required to implement dynamic content, database interactions, user authentication and authorization, CRUD functionality, and secure programming practices.

A demonstration video of 5–10 minutes showcasing the implemented features is a compulsory deliverable.

User Authentication and Roles

The following user roles must be implemented in the real estate system:

Role	Access
Guest	This is the most restrictive role and provides only basic access to the website. Guests can access the following features and pages: • Can browse properties • Registration / Login • About • Contact
Buyer	Users who register as buyers are assigned the Buyer role. Buyers can access all features available to guests and can additionally submit offers and request appointments for properties.
Seller	Users who register as sellers are assigned the Seller role. Sellers can access all features available to guests and can additionally manage their own property listings, appointments, and dashboard.
Administrator	The system must include a default administrator account. Administrators can perform all actions available to sellers and have full access to system management features. Administrators can view all properties, all users (buyers and sellers), all contact enquiries, and access the administrator dashboard

User Registration and Login

Implement user registration and authentication functionality to support the roles described above.

Site Navigation

Modify the site navigation to reflect the authentication state and role of the current user.

Authentication State

- If a user is logged in, display the user's name along with a Logout option.
- If a user is not logged in, display Login and Register options.

Role-Based Navigation

- Navigation links should be displayed according to the user's role and permissions.
- Refer to the User Authentication and Roles section when determining which links should be visible to each user type.
- Ensure navigation is consistent across all pages.

In addition to controlling the visibility of navigation links, access to protected pages must be enforced through authorization mechanisms so that users cannot bypass restrictions by directly entering a page URL.

Landing Page

The landing page serves as the entry point to the real estate platform and is intended to be the first page viewed by visitors.

After authentication, users should be redirected to a role-specific landing page:

- Guest and Buyer: Display the same landing page specified in Assignment 1.
- Seller and Administrator: Redirect users to their dashboard.

Dashboard

Seller Dashboard

The seller dashboard provides summary information relevant to the seller's activity within the system.

It should display data retrieved from the database, including:

- the total number of properties owned by the seller;
- the total number of offers received; and
- the number of pending appointment requests.

Administrator Dashboard

The administrator dashboard provides summary information about the overall system.

It should display data retrieved from the database, including:

- the total number of registered users (buyers and sellers);
- the total number of properties;
- the total number of contact enquiries received; and
- the number of pending approval requests.

Property Listing Page

Implement the same functionality specified in Assignment 1, with property data retrieved from the database rather than static data. The page must support:

- searching,
- filtering,
- sorting, and
- pagination.

Buyers should be able to make offers and request appointments directly from the listing.

Property Details Page

Implement the same functionality specified in Assignment 1, with property information retrieved from the database. The page should display:

- an image gallery,
- property details,
- seller information,
- an Offer button for buyers, and
- an Appointment button for buyers.

Appointment Management

Buyers can request an appointment by selecting a date and time and providing an optional note. Sellers can review appointment requests and either approve or reject them.

Offer Management

Buyers can submit offers by proposing a price for a property. Sellers can respond by accepting, rejecting, or making a counter-offer.

Property Management

This functionality is available to sellers and administrators.

Sellers can:

- add new properties,
- edit existing properties,
- delete properties, and
- change the status of a property.

However, deleting a property listing requires administrator approval.

When creating a property listing, sellers must provide:

- location,
- description,
- property type,
- number of bedrooms,
- number of bathrooms,
- property status,
- expected price, and
- property images.

About Page

Implement the same functionality specified in Assignment 1.

Contact Page

Implement the same functionality specified in Assignment 1. However, submissions from the contact form must be stored in and retrieved from the database as appropriate. The administrator can see the details of contact inquiries received.

Advanced features

The following features are considered **advanced features**. Your group may choose whether to implement these features based on your available time, skills, and project plan.

If you decide to proceed with any advanced features, ensure that you plan ahead and allow sufficient time for implementation and testing. You should prioritise completing all required functionality before attempting advanced features. Implementing additional features should not negatively impact the completeness or quality of your core application.

- **Property Favourites**
Allow buyers to save properties as favourites and manage their preferred property list.
- **Dashboard Analytics**
Provide visual representations of system statistics using charts. Examples include: user distribution (e.g., pie chart showing buyer/seller distribution, offer price trends, the number of contact enquiries, appointments, and offers over time, etc.)
- **Property Status Timeline**
Implement a timeline to track changes in property listing status, such as Available, Pending, and Sold.
- **Notifications**
Implement a notification system where sellers and administrators receive notifications when they log in, informing them of new enquiries, offers, appointment requests, or other relevant activities.

The **Self and Peer Evaluation** must be completed by **Friday, 2 October**.

Submission Method

For the assignment submission, you are required to follow the submission process using both MyLO and usermin. The Readme file, containing essential information, should be submitted specifically to MyLO. The Readme file must include the path to your group's account in usermin, which can be accessed at the following URL: (/groupwork/kit502-groupwork/your-group-number/theassignmentfilename). It is important to note that the last modified date of the assignment files hold significance. To maintain integrity, please refrain from making any modifications to the files after the specified due date.

The Readme file for Assignment 2 should include the login credentials for the system manager, along with any other pertinent information. These credentials will enable the marker to access and evaluate the system manager's privileges and functionalities.

By following these submission instructions and including the necessary details in the Readme file, you will ensure a comprehensive and organised submission process for your assignment.

Assignment Group Account and Database Set up

Each group will be provided with an account specifically for the group assignment. This account will be named "kit502-group-##" where "##" represents your group number. Please make sure to use this account for your collaborative work.

For Assignment 2, you will need to run the Laravel installation command in one of the group member's accounts. After the installation is complete, you should move the Laravel project to the group account using the "mv myApp kit502-group-##" command. This ensures that the project is stored in the designated group account for submission.

Here are the steps you need to follow:

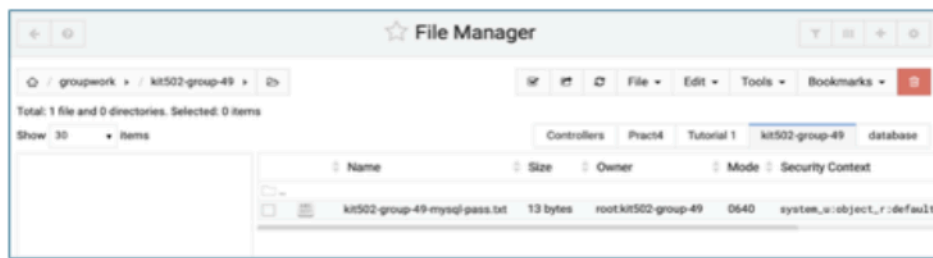
```
cd ~/public_html           //one of members can move the location to public_html
laravel-install             //run this command
mv myApp /groupwork/kit502-group-XX/ //moving myApp to your group folder
```

In order to use a database for Assignment 2, you have two options. If you prefer to use MySQL, you can find the necessary information, including the password, in the file named "kit502-group-##-mysql-pass.txt" within the group account. This file will provide you with the credentials needed to access the MySQL database via PhpMyAdmin.

Alternatively, if you prefer to use phpliteadmin, you can use one of the group member's accounts to create the required tables. This allows you to utilise phpliteadmin for database management during the development process.

By following these instructions, you will be able to set up the necessary account and database environment for Assignment 2. Please ensure that you collaborate effectively and utilise the provided resources to successfully complete your assignment.

For example:



Group Work Evaluation

Meeting Minutes

As this is a group project, you need to provide how you group worked during the due course.

First Step: Task negotiation

Different members of your team will have different strengths and weaknesses. Before you start the task, you must negotiate the roles that you will play on your team and commit to what you will contribute to the assignment. **Your team's expectations must be recorded in an appendix at the end of your report.**

Pay particular attention to these things:

- Set expectations for how, and how often, your group will communicate
- Set expectations for when work is submitted within your group
- Set expectations for the quality of work submitted within your group
- Decide how you will share the workload equally
- See how you can ensure every team member has an opportunity to participate equally
- Set expectations of how you can behave supportively of all team members

Final	Step:	Peer	Evaluation
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After you have completed your Assignment 1 and 2, you will assess your, and your team's contribution to the assignment using a feedback fruit link. Take this seriously. It will affect your marks for this assignment.

To ensure fairness and accuracy in grading, your final mark for Assignment 1 and Assignment 2 will be revised based on the peer and self-evaluation process. It is important that every group member actively participates in this evaluation.

Please carefully complete the peer and self-evaluation forms with any other required form(s), providing qualitative comments and feedback for each member

of your group. This evaluation will contribute to the determination of your final mark. It is crucial that all group members complete the evaluation process, as the final marks will not be released until all evaluations are submitted.

By engaging in peer and self-evaluation, you contribute to the overall assessment of your group's work and help maintain the integrity of the evaluation process.

Rule for Individual mark for group work
Individual marks will be calculated based on the category below.

Category	Range (out of 100)	Ratio
A	81 - 100	1
B	31 - 80	The average peer evaluation mark received
C	0 - 30	0

Your individual mark for the assignment will be calculated based on the average peer evaluation mark received from your group members, as well as your self-evaluation mark. It is important to note that your individual mark may be adjusted based on the evaluation rules outlined above.

To provide you with a clearer understanding of how this calculation works, here are a few examples:

Example for Category A)

Group Mark Received		80
Peer Evaluation mark		
Group member 1	70	
Group member 2	80	
Group member 3	90	
Yourself	100	
Average		85
Individual Mark (Based on Category A): 80×0.85		68

Example for Category B)

Group Mark Received		80
Peer Evaluation mark		
Group member 1	50	
Group member 2	60	
Group member 3	70	
Yourself	100	
Average		70
Individual Mark (Based on Category B): 80×0.70		56

Example for Category C)

Group Mark Received		80
Peer Evaluation mark		
Group member 1	5	
Group member 2	0	
Group member 3	10	
Yourself	100	
Average		28.75
Individual Mark (Based on Category C): 80*0		0

You will only be able to view the group mark for Assignment 1. The individual marks for both Assignment 1 and Assignment 2 will be released at the end of the semester, along with the Assignment 2 marks.

Marking Information

Your submitted work will be evaluated and marked in the usermin environment. It is important to ensure that your assignment functions properly and is compatible with the specified browser version in our lab. If your assignment fails to work correctly or encounters compatibility issues in the lab environment (i.e., your submitted assignment does not work properly in the usermin environment), it may not be marked.

To avoid any potential issues, we recommend thoroughly testing your assignment in the lab environment or using the same browser version to ensure its functionality and compatibility. This will help ensure that your hard work is properly assessed and marked.

Academic Integrity

At the University of Tasmania, academic integrity requires all students to act responsibly, honestly, ethically, and collegially when using, producing, and communicating information with other students and staff members. The University community is committed to upholding the [Statement on Academic Integrity](#).

Breaches of academic integrity such as plagiarism, contract cheating, collusion and so on are counter to the fundamental values of the University. A breach is defined as being when a student: fails to meet the expectations of academic integrity; or seeks to gain, for themselves or for any other person, any academic advantage or advancement to which they or that other person is not entitled; or improperly disadvantages any other member of the University community.

The University and any persons authorised by the University may submit your assessable works to a text matching service, to obtain a report on possible breaches such as plagiarism or contract cheating. Substantiated breaches can result in a range of sanctions which are outlined in the [Student Academic Integrity Ordinance](#).

More information is available from the [Academic Integrity site](#) for students on the Student Portal.

Use of Generative AI



Generative AI Use: The use of Generative AI is **not permitted** for all assessment tasks.

You may use generative AI tools (e.g., ChatGPT) to help you understand concepts and principles or for basic English proofreading, just as you might study with a classmate or seek advice from a friend. However, you must not use AI tools to generate any part of your assessment, including code or written answers, or submit AI-generated content as your own work.

⚠ Submitting AI-generated content as your work constitutes an academic integrity breach.